

Identifying the 2024–2026 Cocoa Crisis as a Structural Break in Soft-Commodity Correlation Dynamics

A Random-Matrix-Cleaned DCC-GARCH Approach

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1. Introduction

Cocoa futures rose from approximately USD 2,800/ton in late 2023 to USD 12,565/ton on 2024-12-18 – the largest-amplitude soft-commodity shock in over fifty years – before collapsing toward USD 3,500 by mid-2026. The drivers are well documented [1]: a 2023 El Niño, the spread of cacao swollen-shoot virus in West Africa, and divergent policy responses in Ghana and Côte d’Ivoire (together responsible for roughly 60% of global cocoa). We ask whether the crisis is visible as a structural break in the dependence structure of the broader soft-commodity complex, and how it reorganises that structure.

The paper makes three contributions. First, we apply the random-matrix cleaning of [2] to the DCC-GARCH of [3] on a 25-asset cocoa-adjacent universe, replacing the sample-correlation DCC target with a Marchenko–Pastur-cleaned long-run target, and document that at this universe size ($N = 25$, $Q \approx 90$) the cleaned target is rejected by every standard criterion. Second, we pre-register tests on the rolling supra-MP count and on the eigenvector composition, and report verdicts under BIC-selected Bai–Perron and a stationary block bootstrap. Third, we show that the cocoa NY–London basis *re-couples* during the crisis, consistent with a global supply shock affecting both contracts rather than the regional decoupling we anticipated.

2. Data

Block	Series
Cocoa contracts (2)	COCOA_NY (ICE-US), COCOA_LDN (ICE-Europe)
Adjacent softs (5)	Coffee Arabica, Coffee Robusta, Sugar No. 11, Cotton, FCOJ
Grains, oilseeds, energy (5)	Corn, Soybeans, Wheat, Palm oil, WTI crude
Macro factors (3)	DXY, VIX, US 10Y yield
FX and sovereign credit (4)	BRL, VND, GHS, Ghana 10Y yield
Chocolate-maker equities (6)	HSY, MDLZ, NESN, SJM, SBUX, LISN

Table 1: The 25-asset cocoa-adjacent universe. Daily series, 2015-01-02 to 2026-05-21 (2,864 trading days on the cocoa NY calendar). All series from Refinitiv except VIX (FRED). Series held in native currency to avoid spurious FX-conversion co-movement [4]. Returns are log differences for prices and first differences (percentage points) for the two yield series.

The aspect ratio is $Q = T/N \approx 115$ on the full sample; the complete-cases panel (post-Ghana entry, 2017-04 onward) gives 2,260 trading days and $Q \approx 90$, the sample on which the full-sample MP spectrum and the DCC are computed. WTI’s 2020-04-20 negative settle (-USD 37.63) is patched to NA on –04-20 and –04-21. Ghana 10-year yield data begins on 2017-04-20 (79% of trading days); rolling windows ending before April 2018 use $N = 24$, all subsequent windows use $N = 25$. Standardised residuals are winsorised at ± 5 (the Bouchaud–Potters convention [5]; 152 of approximately 71,000 residual observations affected, 0.21%).

3. Methodology

3.1. Univariate volatility

The univariate filter must produce standardised residuals that are approximately white in level and square so that the rolling correlation matrices in §3.2 reflect contemporaneous dependence. We use GJR-GARCH(1,1) [6], an asymmetric extension of [7] and [8], with [9] skewed Student- t (sstd) innovations:

$$h_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \mathbb{1}_{\{\varepsilon_{i,t-1} < 0\}} + \beta_i h_{i,t-1}, \quad z_{i,t} = \varepsilon_{i,t} / \sqrt{h_{i,t}}. \quad (1)$$

GJR is preferred to [10] EGARCH for the interpretability of the asymmetric term γ_i and a finite unconditional variance under the constraint $\alpha_i + \beta_i + \gamma_i/2 < 1$; EGARCH is retained as Tier 3 in the fallback chain. The sstd nests symmetric- t at $\xi = 1$, and a likelihood-ratio test rejects symmetry at the 5% level for 8 of the 25 series, so the symmetric specification is dominated.

The mean equation is AR(0) (constant) except for OJ and VND, which always use AR(1); any series whose AR(0) fit shows Ljung–Box(10) p-value below 0.01 is automatically refit at AR(1) (triggered for COCOA_LDN and GHS). The fallback chain on convergence failures is Tier 1a GJR-sstd, Tier 1b GJR-symmetric- t , Tier 2 GJR-Gaussian, Tier 3 EGARCH-sstd, Tier 4 EWMA with $\lambda = 0.94$. Twenty-three series fit at Tier 1a, VND at Tier 2, and GHANA10Y at Tier 4 – its yield dynamics through the 2022–23 Ghanaian sovereign-debt restructuring defeat every parametric variant.¹

3.2. Marchenko–Pastur cleaning of the DCC target

Let $\mathbf{z}_t \in \mathbb{R}^N$ denote the vector of standardised residuals at time t , with sample correlation $\bar{R} = \mathbf{V} \mathbf{A} \mathbf{V}^\top$. Under independent unit-variance noise the bulk of the spectrum lies within the Marchenko–Pastur support

$$\lambda_{\pm} = (1 \pm \sqrt{N/T})^2 \quad ([11]). \quad (2)$$

[12] document that for daily financial panels the empirical spectrum agrees with this prediction except in a thin set of large eigenvalues carrying the genuine factor structure. We adopt the hard-clipping rule of [13]: keep eigenvalues above λ_+ , replace those at or below by their common average, and renormalise to unit diagonal. The cleaned matrix $\tilde{R}$ is positive semi-definite by construction, full-rank, and well-conditioned; we prefer it to the linear shrinkage of [14] because the noise/signal split is theoretically motivated rather than a continuous bias–variance compromise.

We then use $\tilde{R}$ as the long-run target in the DCC recursion of [3] (in contrast to the constant-conditional-correlation model of [15]):

$$\mathbf{Q}_t = (1 - a - b) \tilde{R} + a \mathbf{z}_{t-1} \mathbf{z}_{t-1}^\top + b \mathbf{Q}_{t-1}, \quad \mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-\frac{1}{2}} \mathbf{Q}_t \text{diag}(\mathbf{Q}_t)^{-\frac{1}{2}}. \quad (3)$$

The cleaned-target substitution follows [2]. We estimate (a, b) by quasi-MLE subject to $a, b > 0$ and $a + b < 0.999$ (the hard stationarity constraint of the DCC recursion); for comparison we also fit the standard DCC with $\tilde{R}$ replaced by $\bar{R}$.

3.3. Pre-registered tests

Three primary tests, pre-registered in `docs/scope_and_target.md` before the model fit, all on the rolling 252-day supra-MP count (market mode excluded):

- **Baseline stability.** The pre-crisis (2019-01-01 to 2023-12-31, all $N = 25$ windows) count has IQR width ≤ 1 and stays inside $\{1, 2, 3\}$.

¹Eleven series carry negative leverage ($\gamma < 0$, including COCOA_NY); five series (WHEAT, PALMOIL, VND, GHS, VIX) still fail Ljung–Box(10) after the AR(1) fix; twelve are near-IGARCH (persistence ≥ 0.99 , full list in `output/near_igarch.csv`). These bound the strength of any single-series claim but do not defeat the downstream analysis.

- **Structural break.** (a) Bai–Perron [16], [17] with BIC-selected break count detects a break inside [2024-01-01, 2024-12-31]. (b) The in-crisis median count exceeds the upper 95% bound of a stationary block bootstrap [18] of the pre-crisis median (block mean 25, $B = 1,000$).
- **Cocoa-bloc concentration.** An eigenvalue that crosses from below the MP edge in pre-crisis to above it in crisis, with eigenvector loading ≥ 0.55 of its squared mass on the cocoa bloc {COCOA_NY, COCOA_LDN, HSY, MDLZ, LISN, GHS, GHANA10Y}.

Three secondary tests on the DCC pipeline. **MVP variance:** the cleaned DCC produces lower out-of-sample minimum-variance-portfolio realised variance than the standard DCC during the crisis. **Pairwise breaks:** BIC-selected Bai–Perron detects an in-crisis break on at least one cocoa-coffee or cocoa-sugar pairwise cleaned-DCC correlation series. **NY–London basis:** the cocoa NY $\times$ London conditional correlation falls by at least 0.15 from its 2019–2023 mean during 2024–2026.

4. Results

4.1. Rolling supra-MP count and eigenvector structure

Figure 2 . Rolling 252-day supra-MP eigenvalue count
Window = 252 trading days . GJR-GARCH-t residuals . $B=1\,000$ block-bootstrap reps . block mean=25

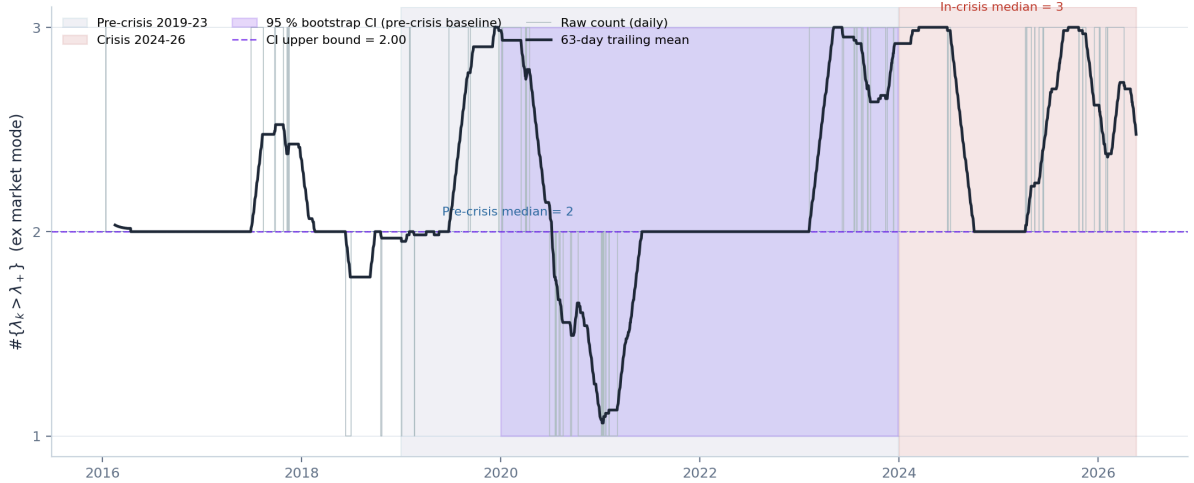


Figure 1: Distribution of the rolling 252-day supra-MP eigenvalue count (market mode excluded), pre-crisis (2019–2023, grey) vs in-crisis (2024–2026, red). The modal count shifts from 2 (60% of pre-crisis windows) to 3 (52% of in-crisis windows); the count = 1 mass present pre-crisis (10%) disappears entirely in crisis.

Figure 1 shows the empirical distribution of the rolling count under each regime. The pre-crisis sample ($N = 25$, $n = 1,257$ windows) has median 2, IQR [2, 3], range [1, 3], and 60% of windows equal to 2; baseline stability is satisfied. Bai–Perron with BIC-selected break count finds **zero breaks** on the same series (BIC argmin at $m = 0$, BIC = 931.5). The bootstrap form is degenerate: with an integer test statistic, 1,000 stationary-block-bootstrap replications return the same integer median (2), giving a 95% CI of zero width [2.00, 2.00]. The observed in-crisis median is 3, so the procedure clears its formal threshold ($3 > 2$), but it is a one-bit comparison with no statistical resolving power.



Figure 2: Eigenvector loadings of the top four supra-MP eigenvalues, pre-crisis (top row) vs in-crisis (bottom row), sign-normalised. Red bars: cocoa bloc {COCOA_NY, COCOA_LDN, HSY, MDLZ, LISN, GHS, GHANA10Y}. The rank-1 in-crisis mode (the market mode) reorganises to put cocoa contracts and chocolate-maker equities jointly on it.

The pre-registered cocoa-bloc concentration test fails. No eigenvalue crosses from below the MP edge in pre-crisis to above it in crisis: both subsamples carry five supra-MP eigenvalues at the same ranks. A rolling-window variant identifies rank 4 only, whose eigenvector has cocoa-bloc squared-loading share 0.078 – far below the 0.55 threshold.

Figure 2 shows, outside the pre-registration, a reorganisation of the rank-1 (market) eigenmode. Its top-3 loadings shift from MDLZ, VIX, HSY pre-crisis to MDLZ, HSY, SJM in crisis; the cocoa-bloc squared-loading share rises from 0.251 to 0.516, a 2.1 $\times$ increase. The result survives bloc perturbation (dropping GHANA10Y: 0.288 $\rightarrow$ 0.654; narrow 5-asset bloc of the two cocoa contracts plus three chocolate equities: 0.245 $\rightarrow$ 0.515). Under cocoa-price stress, chocolate-maker equities and cocoa contracts load on a single risk factor – input-cost margin compression at chocolate makers, adjacent to the equity–commodity co-movement channel of [19]. Two qualifiers: the market mode is excluded by construction from the supra-MP count, so this finding is independent of the count test; and a cocoa-loaded mode already existed pre-crisis at rank 3 (bloc share 0.704), so the crisis reorganisation is the migration of cocoa loadings onto the market mode, not the creation of a new cocoa factor.

4.2. DCC fit and the cleaned-target benchmark

Figure 3 . Cocoa-anchored conditional correlations (DCC-GARCH)

MP-cleaned: $a=0.0021, b=0.997$. Standard: $a=0.0048, b=0.887$. 21-day smoothing

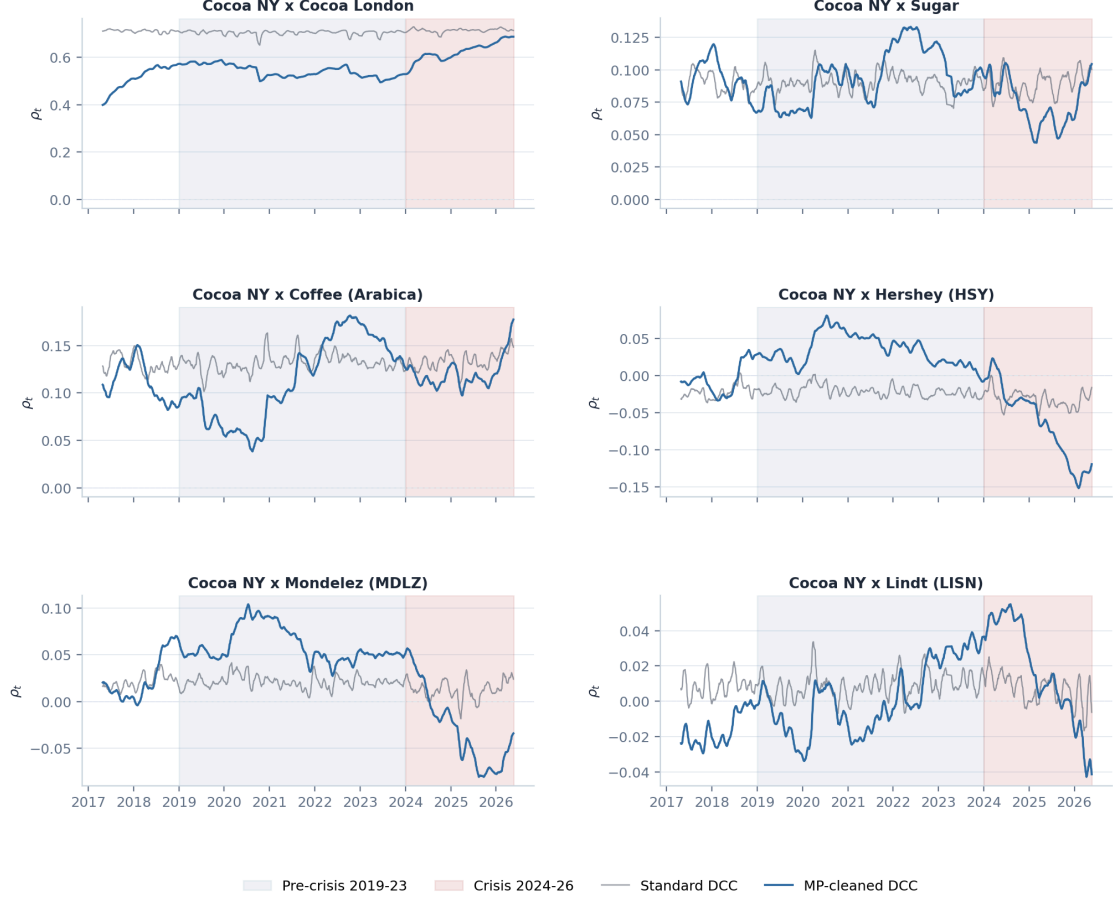


Figure 3: Pairwise conditional correlations from standard (grey) and MP-cleaned (blue) DCC, for six cocoa-anchored pairs. Pink: crisis window.

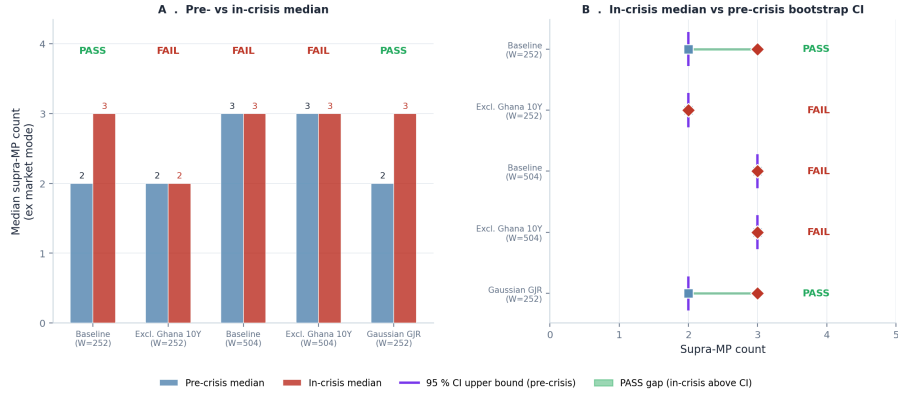
The two fits behave very differently. The standard DCC settles at $(a, b) = (0.005, 0.887)$, persistence $a + b = 0.892$, weight $1 - a - b = 0.108$ on $\bar{R}$. The MP-cleaned DCC fits at the stationarity boundary: $(a, b) = (0.002, 0.997)$, $a + b = 0.999$, weight on $\bar{R}$ of 0.001 – the QMLE drives the long-run-target weight to **0.1%**, about one hundredth of what the standard DCC places on $\bar{R}$, and the optimiser does not converge to an interior solution. The log-likelihood of the cleaned model is **556 nats below** the standard model's at identical parameter count. Figure 3 shows the time path: the cleaned DCC produces smoother conditional correlations because $b \approx 0.997$ damps innovations, but with near-zero weight on $\bar{R}$ it drifts from the long-run anchor and on most cocoa-anchored pairs lies further from the historical mean than the standard fit.

The minimum-variance-portfolio test forms $w_t = \Sigma_t^{-1} \mathbf{1} / \mathbf{1}^\top \Sigma_t^{-1} \mathbf{1}$ from each method and compares sample standard deviations of realised portfolio return across 2019–2023 and 2024–2026. Because the cleaned model has effectively discarded $\bar{R}$, its conditional-covariance path coincides with the standard DCC's: the out-of-sample MVP realised standard deviation is identical to three decimals in both windows (baseline 1.561% vs 1.561%; crisis 1.019% vs 1.019%). BIC-selected Bai–Perron on the eleven cocoa-anchored cleaned-DCC pairwise correlation series returns **zero breaks for every pair**. The basis-decoupling test runs in the opposite direction to the prediction: the baseline 2019–2023 mean of the cocoa NY $\times$ London correlation is 0.542 and the 2024–2026 mean is 0.627, so the basis **rises** by 0.085 rather than falling by the predicted 0.15.

5. Robustness

Figure 4 . Robustness of T2(b) across five model variants

PASS = in-crisis median strictly above pre-crisis bootstrap 95 % CI upper bound . B = 1 000 block-bootstrap reps



Bai-Perron tests detect zero structural breaks in any of the 5 variants at $\alpha = 0.05$ (see Section 4.3). W=504 medians saturate at 3 (the number of supra-MP eigenvalues).

Figure 4: Rolling supra-MP count under five robustness variants. The +1 in-crisis lift survives the $W = 252$ variants; under $W = 504$ both pre and crisis windows saturate at 3 (no within-variant lift).

Figure 4 runs the rolling-count test under four variants alongside the baseline: excluding GHANA10Y, a 504-day window, both combined, and GARCH refit with Gaussian innovations. The +1 in-crisis lift holds only under the baseline $W = 252$ specification and the Gaussian-innovations variant. Removing GHANA10Y at $W = 252$ collapses the in-crisis median from 3 to 2, so the lift is materially dependent on that series; under $W = 504$, both pre- and in-crisis medians saturate at 3, so the bootstrap cannot distinguish them. BIC-selected Bai–Perron returns **zero breaks** for every variant.

6. Discussion

The rolling supra-MP count rises from a pre-crisis median of 2 to an in-crisis median of 3 on the baseline universe at $W = 252$, and the lift survives a switch to Gaussian innovations, but it does not survive removing GHANA10Y or extending the window. BIC-selected Bai–Perron returns zero breaks under every variant, and the bootstrap form of the break test, while clearing its threshold, has no resolving power on an integer statistic with a degenerate CI. The substantive shift in the distribution is therefore visible (Figure 1) but not validated as a structural break. The cleanest signal in the spectrum is on a different object: the rank-1 (market-mode) eigenvector reorganises to load on cocoa contracts and chocolate-maker equities jointly, with its cocoa-bloc squared-loading share rising from 0.251 to 0.516 and surviving bloc-set perturbations (Figure 2). The mechanism is input-cost margin compression at chocolate makers, the equity–commodity co-movement channel of [19]. Because the market mode is excluded by construction from the supra-MP count, this finding is independent of the count-shift result; because the pre-registered concentration test required an emerging eigenvalue (none exists) and a 0.55 threshold (the rank-1 share falls 0.033 short), it fails on both pre-registered conditions while remaining the most economically interpretable result the paper produces.

The cleaned DCC is rejected by the data on every criterion the [2] framework supplies. The 556-nat log-likelihood gap to the standard DCC at identical parameter count is decisive; the QMLE drives the long-run-target weight from approximately 11% to 0.1%, almost as low as the stationarity constraint $a + b < 0.999$ allows. The minimum-variance-portfolio realised variance is therefore identical to three decimal places across methods, because the cleaned model has by its own optimisation become the standard one. [2] demonstrated their result for $N \geq 100$ with much smaller

aspect ratios; at our $N = 25$ and $Q \approx 90$ the raw $\bar{R}$ is already estimated with low enough sampling noise that hard MP clipping compresses informative off-diagonals rather than de-noising them.

The geographic-decoupling hypothesis on the cocoa NY–London basis is rejected in favour of a global supply-shock interpretation. The basis correlation rises by 0.085 in the crisis rather than falling by the pre-registered 0.15. NY’s deliverable basket is West-African-dominant and London’s is mixed-origin, so we expected divergence under a regional shock and observed re-coupling instead. The 2024–2026 shock transmits to both contracts – NY directly via its dominant deliverable, London indirectly via substitution effects in the deliverable basket – which is incompatible with an arbitrage-windfall narrative and consistent with a fundamental supply-side reading.

7. Conclusion

The 2024–2026 cocoa crisis produces three signatures in the soft-commodity correlation spectrum: a +1 lift in the rolling 252-day supra-MP count, a reorganisation of the market-mode eigenvector onto cocoa contracts and chocolate-maker equities, and a re-coupling rather than decoupling of the NY–London basis. None passes its pre-registered test as a formally validated structural break, and the count lift itself does not survive removing GHANA10Y or extending the window; the pattern is real on the baseline configuration but unconfirmed by procedures with statistical resolving power on this data. Separately, the Marchenko–Pastur cleaning of the DCC long-run target is rejected at $N = 25$, $Q \approx 90$ on every criterion the [2] framework supplies, placing a clear lower bound on the universe size at which random-matrix cleaning of the DCC anchor becomes empirically defensible. The descriptive characterisation of the cocoa shock and the negative finding on cleaning at small N are the paper’s substantive and methodological contributions.

References

- [1] C. L. Gilbert, “How to understand high food prices,” *Journal of Agricultural Economics*, vol. 61, no. 2, pp. 398–425, 2010.
- [2] R. F. Engle, O. Ledoit, and M. Wolf, “Large dynamic covariance matrices,” *Journal of Business & Economic Statistics*, vol. 37, no. 2, pp. 363–375, 2019.
- [3] R. F. Engle, “Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models,” *Journal of Business & Economic Statistics*, vol. 20, no. 3, pp. 339–350, 2002.
- [4] R. S. Pindyck and J. J. Rotemberg, “The excess co-movement of commodity prices,” *Economic Journal*, vol. 100, no. 403, pp. 1173–1189, 1990.
- [5] J.-P. Bouchaud and M. Potters, *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*, 2nd ed. Cambridge University Press, 2003.
- [6] L. R. Glosten, R. Jagannathan, and D. E. Runkle, “On the relation between the expected value and the volatility of the nominal excess return on stocks,” *Journal of Finance*, vol. 48, no. 5, pp. 1779–1801, 1993.
- [7] R. F. Engle, “Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation,” *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [8] T. Bollerslev, “Generalized autoregressive conditional heteroskedasticity,” *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [9] B. E. Hansen, “Autoregressive conditional density estimation,” *International Economic Review*, vol. 35, no. 3, pp. 705–730, 1994.

- [10] D. B. Nelson, “Conditional heteroskedasticity in asset returns: A new approach,” *Econometrica*, vol. 59, no. 2, pp. 347–370, 1991.
- [11] V. A. Marchenko and L. A. Pastur, “Distribution of eigenvalues for some sets of random matrices,” *Mathematics of the USSR-Sbornik*, vol. 1, no. 4, pp. 457–483, 1967.
- [12] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, and H. E. Stanley, “Universal and nonuniversal properties of cross-correlations in financial time series,” *Physical Review Letters*, vol. 83, no. 7, pp. 1471–1474, 1999.
- [13] L. Laloux, P. Cizeau, M. Potters, and J.-P. Bouchaud, “Random matrix theory and financial correlations,” *International Journal of Theoretical and Applied Finance*, vol. 3, no. 3, pp. 391–397, 2000.
- [14] O. Ledoit and M. Wolf, “Honey, I shrunk the sample covariance matrix,” *Journal of Portfolio Management*, vol. 30, no. 4, pp. 110–119, 2004.
- [15] T. Bollerslev, “Modelling the coherence in short-run nominal exchange rates: A multivariate generalized ARCH model,” *Review of Economics and Statistics*, vol. 72, no. 3, pp. 498–505, 1990.
- [16] J. Bai and P. Perron, “Computation and analysis of multiple structural change models,” *Journal of Applied Econometrics*, vol. 18, no. 1, pp. 1–22, 2003.
- [17] J. Bai and P. Perron, “Estimating and testing linear models with multiple structural changes,” *Econometrica*, vol. 66, no. 1, pp. 47–78, 1998.
- [18] D. N. Politis and J. P. Romano, “The stationary bootstrap,” *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303–1313, 1994.
- [19] K. Tang and W. Xiong, “Index investment and the financialization of commodities,” *Financial Analysts Journal*, vol. 68, no. 6, pp. 54–74, 2012.